Fully timed native isogeny-fiber relation comparison

2026-10-13 · T-2486 · four non-isomorphic degree-three pairs · 640/640 measured answers verified

The frozen [protocol](PROTOCOL.md) selected prime-order subgroups 103, 439, 1,613 and 6,427 on `E/F_p: y²=x³+1` and its degree-three destination `E′/F_p: y²=x³−27`. Eight public source targets per size were solved by cold source relations, cold transported relations, and source negation rho. Each had five measured rounds and one warmup; a duplicate rho supplied an A/A control. Exact identities and model/generator records are in `identities.tsv`. The map is Vélu's `A=0` formula (Pizzo, Pomerance and Voight, equation (2.7)); all rational image fibers and non-isomorphism checks passed at the four selected fields.

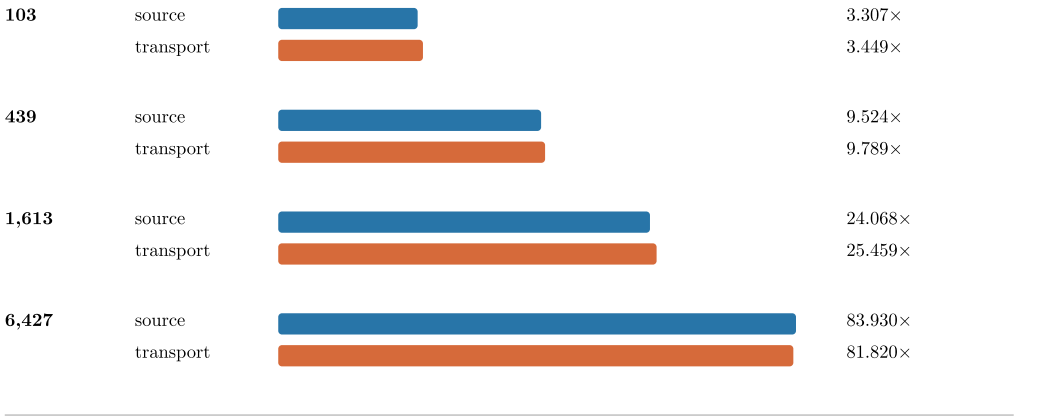
Current-thread CPU time directly covers each arm from its own setup through the verified scalar. It prices factor-base roots, cofactor projections, pair-table work, modular elimination, map evaluation, descent, rho class work and verification. The host was shared Darwin arm64, L0; this is a local complete-cost observation, without wall-clock admission.

r	arm	CPU ms	vs rho	paired 95% interval
103	rho	0.007055	1.000	—
	source	0.023328	3.307	[2.318, 4.953]
	transport	0.024330	3.449	[2.452, 5.014]
439	rho	0.007010	1.000	—
	source	0.066758	9.524	[8.087, 11.382]
	transport	0.068613	9.789	[7.470, 12.272]
1,613	rho	0.014116	1.000	—
	source	0.339743	24.068	[20.222, 28.456]
	transport	0.359370	25.459	[22.261, 28.871]
6,427	rho	0.020155	1.000	—
	source	1.691627	83.930	[67.667, 104.252]
	transport	1.649107	81.820	[69.169, 96.622]

Comparison and retained controls

Complete cold thread-CPU work versus matched negation rho

Four non-isomorphic degree-three pairs; eight public targets and five measured rounds per size. Bars use a $\log_2$ axis: one bit of ratio is 0.61 inches. Exact values and paired target intervals are in `summary-v3.tsv`.



■ source relations · ■ destination relations. All 320 V3 relation records recovered and independently checked their scalars. Rho is 1× on this axis. A/A thread-CPU ratios were 0.810–0.918; shared-host wall timing failed its noise control. This is an L0 local CPU observation and is not an admitted wall-clock speed claim.

Transport/source CPU ratios were 1.043, 1.028, 1.058 and 0.975, with all paired intervals crossing 1. The rho A/A CPU ratios in V3 were 0.810, 0.918, 0.819 and 0.824, within the frozen 0.8–1.25 gate but large enough to limit the interpretation of those small source/destination differences. The generic two-class step floor is `sqrt(pi/4)=0.886227` in the separate counted-group `S` unit. Its phase counts and boundary ratios appear in `summary-v3.tsv`; they omit native operations that the CPU unit includes.

Why the run versions are distinct

Duplicate rho control across the retained run versions

Each cell is the full-work duplicate-rho / matched-rho ratio. Both copies performed the same counted group operations on the same target and seed. V1 uses elapsed wall time; V2 and V3 use current-thread CPU time. V2/V3's frozen admissible A/A band is 0.8–1.25.

subgroup r	V1 wall	V2 CPU	V3 CPU
103	420.987×	0.827×	0.810×
439	0.771×	0.924×	0.918×
1,613	2.071×	0.754×	0.819×
6,427	5.690×	0.817×	0.824×

Pink marks a failed or visibly unstable control. Green marks the frozen CPU ratio gate only. The residual 8–19% A/A CPU imbalance limits the interpretation of the smaller source-versus-destination differences. Source: `summary-v1.tsv`, `summary-v2.tsv`, `summary-v3.tsv` and retained raw records beside them.

V1′s wall A/A ratios were 420.987, 0.771, 2.071 and 5.690; its wall comparison is censored by scheduling. V2 added current-thread CPU time, but its order-1,613 A/A ratio 0.754 missed the frozen gate. V3 balanced the arm order and passed that gate. All three versions retained their raw rows, source snapshots, run logs, independent relation replay and deterministic replay receipts. The checker independently reconstructed 384 relation records per version, including warmups, with its own point arithmetic and modular inverse.

The destination arm evaluated the isogeny twice per target and rebuilt its factor base and pair table per solve. Process peak RSS was 25.9–27.2 MiB; it is not an arm-specific memory value. Research preflight took 2.858 s, 0.0014 s, 0.0073 s and 0.711 s in size order, reported separately from the public-input single-target solve. Candidate band search was not timed. There was no configured isolab client or independent host replay. L2 wall admission and a portable ecbench counted price remain open. The linked `REPORT.md`, raw files and `ARTIFACTS.sha256` give the complete evidence and exact commands.

The pinned Lean module proves two finite sixth-power exclusions and four coprimality checks. The two larger exclusions and the elliptic-curve classification/map bridge remain open formalization obligations; the failed full-module attempts are retained beside the partial build.